24. An improved variety of transgenic basmati rice (AIPMT,2010)
    a) Does not require chemical fertilisers and growth hormones
    b) **Gives high yield and is rich in vitamin A**
    c) Is completely resistant to all insect pests and diseases of paddy
    d) Gives high yield but no characteristic aroma

25) Consumption of which one of the following foods prevent the kind of blindness associated with vitamin A deficiency? (AIPMT 2012)
    a) Flavr Savr b) Canola
    c) **Golden rice** d) Bt brinjal

26. A protoplast is a cell (NEET 2016)
    a) undergoing division
    b)without cell wall
    c) without plasma membrane
    d) without nucleus.

27. A technique of micropropagation is (NEET 2015)
    a) Protoplast fusion
    b) embryo rescue
    c) somatic hybridization
    d) somatic embryogenesis

28. To obtain virus-free healthy plants from a diseased one by tissue culture technique, which part/parts of the diseased plant will be taken? (AIPMT 2014)
    a) Apical meristem only
    b) Palisade parenchyma
    c) **Both apical and axillary meristems**
    d) Epidermis only.

29. Cellular totipotency was demonstrated by (AIPMT 1991)
    a) Theodore Schwann
    b) A.V. Leeuwenhoek
    c) F.C. Steward
    d) Robert Hooke

30. Tissue culture technique can produce infinite number of new plants from a small parental tissue. The economic importance of the technique is raising. (Karnataka NEET 2013)
    a) **genetically uniform population identical to the original parent.**
    b) homozygous diploid plants
    c) new species
    d) variants through picking up somaclonal variations

31. Which of the following statements is not true about somatic embryogenesis? (Karnataka NEET 2013).
    a. The pattern of development of a somatic embryo is comparable to that of a zygotic embryo.
    b) **Somatic embryos can develop from microspores.**
    c) Somatic embryo is induced usually by an auxin such as 2, 4-D.
    d) A somatic embryo develops from a somatic cell.

32. Which one of the following is a case of wrong matching? (AIPMT 2012)
    a) Somatic - Fusion of two diverse hybridization cells
    b) Vector DNA - Site for tRNA synthesis
    c) Micropropagation - *in vitro* production of plants in large numbers
    d) Callus - Unorganised mass of cells produced in tissue culture.

33. Polyethylene glycol method is used for (AIPMT 2010)
    a) biodiesel production
    b) seedless fruit production
    c) energy production from sewage
    d) gene transfer without a vector.

34. Somaclones are obtained by (AIPMT 2009)
    a) Plant breeding
    b) Irradiation
    c) genetic engineering
    d) tissue culture.

TN_GOVT_BOTANY_XII_PAGES_216-279 CH 10.indd 271 28-12-2022 15:39:23